

Analysis of an X17-like Search

[Técnicas de análisis y simulación en física nuclear y de partículas 2025-2026
(Máster Universitario en Física) / Yassid Ayyad]

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Overview

In this exercise, you will simulate and analyze a simplified nuclear experiment sensitive to a hypothetical light boson with mass near $17 \text{ MeV}/c^2$, often referred to as the *X17*. The purpose of the exercise is not only to study a possible beyond-Standard-Model signal, but also to learn the standard workflow used in high-energy and nuclear physics: Monte Carlo event generation, detector simulation with Geant4, data storage in ROOT format, and offline analysis with a small ROOT-based framework.

You will be provided with angular distributions for the emitted positron and electron. Using these distributions, you will generate events, simulate their passage through a detector, reconstruct the relevant observables, and compare the reconstructed opening-angle distribution with the expected physics model.

Introduction to the X17

The X17 is a hypothetical neutral boson proposed to explain anomalies reported in the angular correlation of electron–positron pairs emitted in some nuclear transitions, especially in studies of ^8Be and later related nuclei. In the X17 interpretation, the nuclear transition proceeds through an intermediate boson that decays into an e^+e^- pair, producing a characteristic distortion in the opening-angle distribution relative to the smooth expectation from internal pair creation. The topic remains under active discussion because different experiments have investigated the anomaly with different detectors, acceptances, resolutions, and reaction channels.

Objectives

- Translate a physics model into a Monte Carlo event generator using supplied angular distributions.
- Implement a simplified detector geometry and particle transport in Geant4.
- Record event-level information in ROOT ntuples or histograms.
- Build a ROOT-based analysis chain to read, select, histogram, and fit events.
- Reconstruct the e^+e^- opening angle and compare it with the generator-level input.
- Quantify the impact of acceptance, resolution, and event-selection cuts.
- Present the results in a clear format.

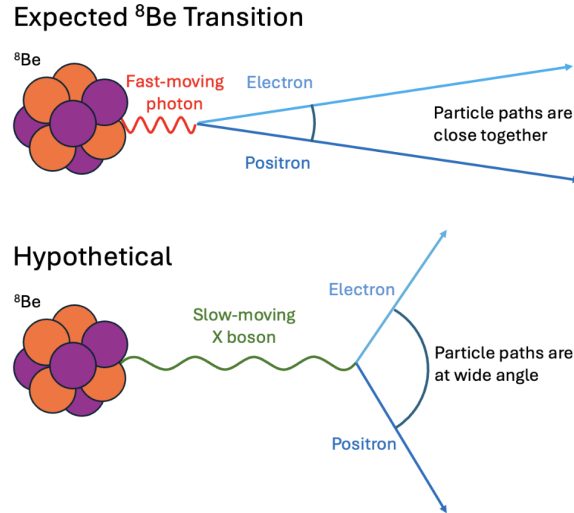


Figure 1: Graphical representation of the ${}^7\text{Li}(p, e^- e^+) {}^8\text{Be}$ process: the creation of the massive X17 boson decaying into an $e^- e^+$ pair is deduced by the excess of $e^- e^+$ pairs at a large relative angle. The main background is due to the internal pair conversion (IPC) of virtual photons.

Problem Statement

The main task is to simulate a simplified experiment searching for an X17-like signal in nuclear pair emission. The input to the simulation will be the angular distributions of the emitted positron and electron, which you must use to generate event kinematics.

You will then build a Geant4-based detector simulation, propagate the particles through the detector geometry, and store the relevant observables in a ROOT file. Finally, you will write a small ROOT analysis framework to reconstruct the opening-angle distribution, compare the reconstructed data with the expected model, and study the effect of detector response on the observable.

The main goal is to understand how a realistic detector modifies the physical signal through finite resolution, geometric acceptance, thresholds, and analysis selections.

Tasks

1. **Event generation.** Implement the generation of electron and positron directions using the provided angular distributions.
2. **Detector simulation.** Build a simplified Geant4 geometry representing the target and detector system. **Choose the technology and geometry you consider more convenient.**
3. **Transport and hits.** Simulate the transport of particles through the detector and record the relevant hit or track information.
4. **ROOT output.** Store the relevant observables in a ROOT file using ntuples and/or histograms.
5. **Offline analysis.** Write a ROOT-based analysis that:
 - Reads the output file.
 - Applies basic quality cuts.

- Reconstructs the opening angle.
6. **Detector effects.** Study the effect of acceptance and finite resolution on the angular distribution.
 7. **Final report.** Produce a short report with plots, code structure, and physics interpretation.

Optional Extensions

- Compare several detector geometries.
- Include background events.
- Study how the result changes when the angular resolution is degraded.

Suggested Reference Material

1. Gustavino, C. X17: Status and Perspectives. *Universe* 2024, 10, 285.
<https://doi.org/10.3390/universe10070285>
2. A. J. Krasznahorkay et al., *An Update on the Hypothetical X17 Particle*, arXiv:2409.16300.
3. ROOT documentation on histogram fitting and analysis:
<https://root.cern/manual/fitting/>
4. Geant4 documentation and tutorials on analysis and data persistence:
<https://geant4.web.cern.ch/documentation/>